

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – Enumeration Analysis

COURSE CODE: ITA14 **COURSE NAME:** Ethical Hacking

COURSE OUTCOME COVERED

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

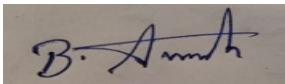
Total Marks: 100

Code of Conduct:

I Avinash (192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink on a light-colored background. The signature appears to be "B. Avinash" written in a cursive, flowing style.

CO3 AT1 SCENARIO BASED ASSESSMENT

Question 1

```
(avinash@kali)-[~]
$ sudo nmap -sS -sV -O -sC scanme.nmap.org
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 20:44 +0530
Stats: 0:00:44 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.65% done; ETC: 20:45 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Device type: general purpose
Running (JUST GUESSING): Linux 5.X (85%)
OS CPE: cpe:/o:linux:linux_kernel:5
Aggressive OS guesses: Linux 5.0 - 5.14 (85%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submi
Nmap done: 1 IP address (1 host up) scanned in 51.38 seconds
```

Question 2

```
(avinash@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.

— 192.168.56.101 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3071ms

(avinash@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:05:55:12 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:01:70:64 brd ff:ff:ff:ff:ff:ff
    inet 10.154.71.4/24 brd 10.154.71.255 scope global dynamic noprefixroute eth1
        valid_lft 2149sec preferred_lft 2149sec
    inet6 2409:40f4:4481:c8d1:5fa8:f6eb:62b8:d0c0/64 scope global temporary dynamic
        valid_lft 6994sec preferred_lft 6994sec
    inet6 2409:40f4:4481:c8d1:a00:27ff:fe01:7064/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6994sec preferred_lft 6994sec
    inet6 fe80::a00:27ff:fe01:7064/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(avinash@kali)-[~]
$ ip route
default via 10.154.71.108 dev eth1 proto dhcp src 10.154.71.4 metric 100
10.154.71.0/24 dev eth1 proto kernel scope link src 10.154.71.4 metric 100

(avinash@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.

— 192.168.56.101 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3048ms
```

Question 3

```
(avinash@kali)-[~]
$ sudo nmap -sS -sV -O -sC -p- 192.168.56.102
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:01 +0530
Stats: 0:01:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 28.96% done; ETC: 21:06 (0:03:09 remaining)
Nmap scan report for 192.168.56.102
Host is up (0.071s latency).
All 65535 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: power-device|firewall|NMAP|router|general purpose|specialized
Running (JUST GUESSING): APC embedded (94%), Cisco ASA 9.X (94%), Cisco embedded (94%), Synology embedded (94%), Microsoft Windows 2000/2003 (93%), ETH Zurich Bluebottle (87%), Tibbo embedded (86%)
OS CPE: cpe:/o:cisco:asa:9.2 cpe:/h:synology:rt1900ac cpe:/o:microsoft:windows_server_2003:sp4 cpe:/o:ethzurich:bluebottle
Aggressive OS guesses: APC Network Management Card 3 (94%), Cisco Adaptive Security Appliance (ASA 9.2) (94%), Cisco Aironet 3800-series NMP (94%), Synology RT1900ac router (94%), Microsoft Windows 2000 SP4 (93%), Microsoft Windows Ser
ver 2003 SP2 (93%), Bluebottle OS (87%), Tibbo DS10xx serial controller, V3.34 (86%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 398.76 seconds

(avinash@kali)-[~]
$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.

— 192.168.56.102 ping statistics —
4 packets transmitted, 0 received, 100% packet loss, time 3063ms

(avinash@kali)-[~]
$ sudo nmap -p- -sV --top-ports 1000 192.168.56.102
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:18 +0530
Nmap scan report for 192.168.56.102
Host is up.
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 202.28 seconds
```

Question 4

```
(avinash@kali)-[~]
└─$ sudo nmap -Pn -sV -O -sC --top-ports 1000 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:38 +0530
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.50% done; ETC: 21:42 (0:03:41 remaining)
Stats: 0:01:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.00% done; ETC: 21:41 (0:02:06 remaining)
Nmap scan report for 192.168.56.103
Host is up (0.038s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: power-device|firewall|WAP|router|general purpose|specialized
Running (JUST GUESSING): APC embedded (94%), Cisco ASA 9.X (94%), Cisco embedded (94%), Synology embedded (94%), Microsoft Windows 2000|2003 (93%), ETH Zurich Bluebottle (87%), Tibbo embedded (86%)
OS CPE: cpe:/o:cisco:asa:9.2 cpe:/h:synology:rt1900ac cpe:/o:microsoft:windows_2000::sp4 cpe:/o:microsoft:windows_server_2003::sp2 cpe:/o:ethzurich:bluebottle
Aggressive OS guesses: APC Network Management Card 3 (94%), Cisco Adaptive Security Appliance (ASA 9.2) (94%), Cisco Aironet 3800-series WAP (94%), Synology RT1900ac router (94%), Microsoft Windows 2000 SP4 (93%), Microsoft Windows Ser
ver 2003 SP2 (93%), Bluebottle OS (87%), Tibbo DS10xx serial controller, V3.34 (86%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 210.14 seconds

(avinash@kali)-[~]
└─$ nmap -Pn -sV -p 80,443,3000,8080 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:42 +0530
Nmap scan report for 192.168.56.103
Host is up.

PORT      STATE SERVICE VERSION
80/tcp    filtered http
443/tcp   filtered https
3000/tcp  filtered ppp
8080/tcp  filtered http-proxy

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.83 seconds

(avinash@kali)-[~]
└─$ ping -c 4 192.168.56.103
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.

--- 192.168.56.103 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3057ms
```

Question 5

```
(avinash@kali)-[~]
└─$ nmap -Pn -p 139,445 -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:45 +0530
Nmap scan report for 192.168.56.102
Host is up.

PORT      STATE SERVICE VERSION
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.83 seconds

(avinash@kali)-[~]
└─$ sudo nmap -Pn -p 139,445 --script smb-enum-shares,smb-enum-users,smb-os-discovery -sV 192.168.56.102
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:46 +0530
Nmap scan report for 192.168.56.102
Host is up.

PORT      STATE SERVICE VERSION
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.85 seconds

(avinash@kali)-[~]
└─$ nmap -Pn -p 139,445 -sV 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 21:47 +0530
Nmap scan report for 192.168.56.102
Host is up.

PORT      STATE SERVICE VERSION
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.82 seconds
```